

POST LAB (PRACTICAL 11)-

Aim -Implementing a Virtual Memory System with Demand Paging where pages are loaded into memory only when needed, optimizing memory usage and minimizing page faults.

CODE-

```
#include <stdio.h>

int main() {
    int pages[50], frame[10];
    int n, f, i, j, k, flag, page_faults = 0, index = 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string: ");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    scanf("%d", &f);

    // Initialize frames to -1
    for(i = 0; i < f; i++) {
        frame[i] = -1;
    }

    printf("\nPage\tFrames\n");

    for(i = 0; i < n; i++) {
        flag = 0;

        // Check if page already in frame
        for(j = 0; j < f; j++) {
            if(frame[j] == pages[i]) {
                flag = 1;
                break;
            }
        }

        // If page not found -> Page Fault
        if(flag == 0) {
            frame[index] = pages[i];
            index = (index + 1) % f; // FIFO replacement
            page_faults++;
        }

        // Print frames
        printf("%d\t", pages[i]);
        for(k = 0; k < f; k++) {
            if(frame[k] != -1)
                printf("%d ", frame[k]);
            else
                printf("- ");
        }
        printf("\n");
    }

    printf("\nTotal Page Faults = %d\n", page_faults);

    return 0;
}
```

OUTPUT-

```
HP@DESKTOP-B4BD73N MINGW64 ~  
$ nano demand_paging.c  
  
HP@DESKTOP-B4BD73N MINGW64 ~  
$ gcc demand_paging.c -o demand  
  
HP@DESKTOP-B4BD73N MINGW64 ~  
$ ./demand  
Enter number of pages: 12  
Enter page reference string: 1 2 3 4 1 2 5 1 2 3 4 5  
Enter number of frames: 3  
  
Page    Frames  
1        1 - -  
2        1 2 -  
3        1 2 3  
4        4 2 3  
1        4 1 3  
2        4 1 2  
5        5 1 2  
1        5 1 2  
2        5 1 2  
3        5 3 2  
4        5 3 4  
5        5 3 4  
  
Total Page Faults = 9
```